

Primitive Affine Columns at a Distinguished Zero Frequency: Fixed–Determinant Projective Plancherel, Structured–Mask Disintegration, and the Q^3 Energy Gate

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July 2026

Abstract

We continue the analysis of the distinguished zero-frequency coefficient left by a matched prime–Möbius cutoff shell. The earlier reduction identifies the required physical amplitude but leaves its two-point Möbius structure and actual row-pair mask entangled. We first return from the abstract bounded-mask interface to the original mask

$$\mathbf{1}_{\ell \neq \ell'} \mathbf{1}_{|\ell d - \ell' e| > QX^{-\kappa_0}} \mathbf{1}_{(d,e) \leq X^{\kappa_0}}.$$

After fixing the opposite row, its gcd condition has the exact expansion

$$\mathbf{1}_{(d,e) \leq G} = \sum_{v|d, v|e} \lambda_G(v), \quad \sum_{v|e} |\lambda_G(v)| \leq 3^{\omega(e)} = X^{o(1)}.$$

The near-diagonal deletion cuts the remaining row interval into at most two pieces. Opening one ultra divisor and reflecting its complementary factor therefore decomposes each physical column, with only $X^{o(1)}$ coefficient mass, into primitive affine correlations

$$\sum_{r \in I} \mu(\tilde{d}_0 + sr) \mu(\tilde{u}_0 + \ell jvr) W(r), \quad s\tilde{u}_0 - \ell jv\tilde{d}_0 = h, \quad |I| \ll 1 + \frac{D}{sv}.$$

This is a columnwise theorem for the actual mask, not a low-rank theorem for an arbitrary bounded mask.

For the arithmetic core, every equation $sU - aD = h$ with $(a, s) = 1$ and $(as, h) = 1$ is put into an integral SL_2 -normal form

$$U_t = hx + at, \quad D_t = hy + st, \quad sx - ay = 1.$$

It yields the exact identities

$$(D_t, U_t) = (D_t, h) = (U_t, h) = (h, t)$$

and converts the primitive Möbius product into the reducible quadratic sequence $\mu(D_t U_t)$, whose discriminant is h^2 . With a balanced Bézout origin we prove uniformly for $K = \max(a, s)$ that

$$\sum_{\substack{T < t \leq 2T \\ (t, h) = 1}} \mu^2(D_t) \mu^2(U_t) = \mathfrak{S}_{\text{sf}}(a, s; h) T + O_h \left(\frac{T}{\log(2T)} + \sqrt{KT} \right),$$

where the Euler product is explicit and bounded below by a positive constant depending only on h . This controls squarefree support, not Möbius signs.

We then prove an exact fixed-determinant projective Plancherel identity. For a prime q , $h \neq 0 \pmod{q}$, and functions $f, g : \mathbb{F}_q \rightarrow \mathbb{C}$, let

$$\mathcal{C}_{a,s}^{(h)}(f, g) = \sum_{D \in \mathbb{F}_q} f(D)g(s^{-1}(aD + h)).$$

Writing $f^\circ = f - \bar{f}$ and $g^\circ = g - \bar{g}$, one has

$$\sum_{a,s \neq 0} \left| \mathcal{C}_{a,s}^{(h)} - q\bar{f}\bar{g} \right|^2 = q\|f^\circ\|_2^2\|g^\circ\|_2^2 - \sum_{b \neq 0} \left| \sum_D f^\circ(D)g^\circ(bD) \right|^2.$$

Consequently the finite Möbius model has $o(q)$ correlation for all but $O(q^{2-2\eta})$ nonzero coefficient pairs (a, s) , for every $0 < \eta < 1/2$. Complete $a = \ell j$ orbits preserve this energy exactly, even if ℓ is restricted to primes.

Finally, the two exact polarizations of the physical shell define column energies E_L, E_R . We prove the conditional transfer

$$E_L + E_R \ll X^{o(1)}Q^3 \implies |\mathcal{S}_{\text{sh}}^{\text{all}}| \ll X^{o(1)}Q^2.$$

At $Q = X^{267/400+o(1)}$, this is precisely the $J^{-1} = X^{-133/400+o(1)}$ amplitude saving required by the preceding zero-frequency gate. The Q^3 estimate is not proved here. A single-fiber square-root strategy falls short by 49727/210000, and a single finite-field modulus cannot simultaneously provide integer no-wrap and physical slope coverage. Thus the remaining target is a restricted physical projective-energy estimate, not pointwise two-point Chowla. No prime-pair asymptotic, twin-prime theorem, positivity result, or breach of the parity barrier is claimed.

Keywords: primitive Möbius tail; affine Möbius correlation; fixed determinant; projective Plancherel; squarefree sieve; structured row mask; zero frequency; parity barrier.

MSC 2020: Primary 11N35, 11N37; Secondary 11B25, 11L07, 11T23.

1 Introduction

The purpose of this paper is to turn one exceptional Fourier coefficient into a concrete arithmetic average. TPC-32 proves determinant-frequency dispersion for a matched prime–Möbius cutoff shell, but the physical problem selects the coefficient at frequency zero [14]. Ordinary frequencies are small outside a power-small exceptional set; membership of the distinguished coefficient in that set is not decided by Parseval. The same paper shows that support size, ℓ^1 -mass, ℓ^2 -energy, and even perfect cancellation at every nonzero frequency do not force cancellation at zero. Additional arithmetic structure is required.

There are two logically separate questions. First, what is the exact arithmetic sequence inside a zero-frequency column after the actual row-pair mask is restored? Second, can averaging over the physical family of those sequences replace a pointwise two-point Möbius theorem? The present paper answers the first question and proves a finite-field model for the second. It does not complete the final transfer to the integer physical packet.

1.1 From a bounded mask to the actual column

The abstract interface of TPC-25–TPC-32 deliberately retains the row-pair mask as an arbitrary bounded joint multiplier [11, 14]. That is the correct interface for the earlier norm estimates, but it is too broad for an affine-column theorem: an arbitrary entrywise bounded matrix can

align with any prescribed kernel. We therefore return to the actual mask inherited from TPC-18 and recorded in TPC-21 [9],

$$\mathbf{m}((\ell, d), (\ell', e)) = \mathbf{1}_{\ell \neq \ell'} \mathbf{1}_{|\ell d - \ell' e| > Q X^{-\kappa_0}} \mathbf{1}_{(d, e) \leq X^{\kappa_0}}. \quad (1)$$

Here d, e are squarefree and lie on one opened dyadic scale. For a fixed opposite row (ℓ', e) , the last indicator has a divisor expansion of total coefficient mass $3^{\omega(e)} = X^{o(1)}$, while the middle indicator merely removes one interval from the d -line. After one ultra divisor is reflected, the remaining inner variable lies on

$$\tilde{d}_r = \tilde{d}_0 + sr, \quad \tilde{u}_r = \tilde{u}_0 + \ell j v r, \quad s \tilde{u}_0 - \ell j v \tilde{d}_0 = h. \quad (2)$$

Thus the actual mask has small *columnwise* expansion complexity. This does not say that the entire mask matrix has small projective or Schur norm.

1.2 Exact arithmetic of a fixed-determinant line

The equation

$$sU - aD = h \quad (3)$$

is not merely a nonproportional pair of affine forms. Under the physical primitivity conditions $(a, s) = 1$ and $(as, h) = 1$, it belongs to one unimodular orbit. Choosing $sx - ay = 1$ gives

$$\begin{pmatrix} U_t \\ D_t \end{pmatrix} = \begin{pmatrix} x & a \\ y & s \end{pmatrix} \begin{pmatrix} h \\ t \end{pmatrix}.$$

The determinant-one matrix transfers all common divisors back to the two coordinates (h, t) . In particular, the primitive part of the two-sign sequence is one Möbius value on the reducible quadratic

$$D_t U_t = ast^2 + h(sx + ay)t + h^2 xy, \quad \text{disc}(D_t U_t) = h^2. \quad (4)$$

This reformulation does not prove signed cancellation. We do prove a uniform squarefree-support theorem for a balanced Bézout origin. Its main term has a positive Euler product uniformly in the growing slopes when $\max(a, s) = o(T)$ by a fixed power.

1.3 Projective averaging at fixed determinant

The central unconditional average theorem is finite and exact. Over a prime field, the change of variables

$$(a, s) \longmapsto (b, c) = (a/s, h/s) \quad (5)$$

is a bijection of $(\mathbb{F}_q^\times)^2$. Translation Parseval in c , followed by summation over the slope b , gives an exact second moment for every fixed nonzero determinant h . Outside $O(q^{2-2\eta})$ coefficient pairs, it yields an $O(q^{1/2+\eta})$ centered fluctuation and hence $o(q)$ total correlation in the finite Möbius model. Moreover, on a complete residue orbit the restriction $a = \ell j$ loses no energy: multiplication by every fixed nonzero ℓ , prime or not, permutes $j \in \mathbb{F}_q^\times$.

This theorem rotates the earlier “almost every determinant frequency” statement into “fixed determinant, almost every nonzero coefficient pair.” The average runs over both $b = a/s$ and the affine offset $c = h/s$; by itself it is not an almost-every- b statement. A direct physical consequence still fails for a precise reason. Integer no-wrap needs a modulus on the ambient determinant scale, approximately X , whereas complete coverage of the orbit parameter is available only for moduli at most $J = X^{133/400+o(1)}$. A single modulus cannot satisfy both requirements.

Table 1: Logical status of the TPC-33 interfaces.

Statement	Status	What it controls
$SL_2(\mathbb{Z})$ affine normal form and gcd identities	unconditional	every primitive line $sU - aD = h$
Balanced-origin squarefree energy	unconditional	support, not signs
Actual-mask divisor expansion	unconditional	one fixed physical column
Fixed- h projective Plancherel	unconditional	complete two-parameter finite family
Finite Möbius	unconditional	modular model, not integer shell
almost-all-coefficient-pair bound		
Q^3 column energy $\Rightarrow Q^2$ shell	conditional transfer	distinguished physical zero mode
Restricted physical Q^3 energy	open	next analytic gate

1.4 The exact conditional gate

Let C_m denote the ultra increment and H_m the half-prefix from the exact polarization in TPC-32. The full matched shell is

$$\mathcal{K}_{\alpha,\gamma}^{\text{sh}}(j) = C_{m_\alpha}(j)H_{m_\gamma}(j) + H_{m_\alpha}(j)C_{m_\gamma}(j). \quad (6)$$

Keeping the opposite row outside each column defines two nonnegative column energies E_L, E_R . The inherited one-row estimate $\|\gamma^{(i)}\|_2^2 \ll Q \log(2L)$ gives the rigorous implication

$$E_L + E_R \ll X^{o(1)}Q^3 \implies |\mathcal{S}_{\text{sh}}^{\text{all}}| \ll X^{o(1)}Q^2. \quad (7)$$

The left side is a new analytic input, not a consequence of the row-energy bound. Separately estimating the two polarizations is sufficient but possibly stronger than estimating their scalar sum.

At the source-compatible endpoint of TPC-28,

$$Q = X^{267/400+o(1)}, \quad J = X^{133/400+o(1)}, \quad JQ \asymp X, \quad (8)$$

and therefore $Q^2 = (JQ^2)J^{-1}$. The right side of (7) is exactly the amplitude budget of the preceding zero-frequency gate. It is important not to restate this as a proved flatness ratio: no matching lower bound for the determinant-cell energy is known.

1.5 Result ledger and scope

The logical status of the main statements is summarized in [table 1](#).

The new contribution should be read as an exact synthesis tailored to the TPC residual: the structured physical disintegration, the fixed-determinant projective identity with its prime-product lift, and the quantified energy splice. Basic finite Fourier orthogonality and squarefree sieving have a large classical literature, so no priority claim is made for those ingredients in isolation. Nor do the results imply a Hardy–Littlewood main term, positivity of a prime-pair correlation, infinitely many twin primes, or a general resolution of the sieve parity problem.

The paper is organized as follows. [Section 2](#) fixes the matched shell and the actual mask. [Section 3](#) proves the integral normal form, and [section 4](#) proves the support theorem. [Section 5](#) performs the physical column disintegration. [Section 6](#) proves the finite-field identity, and [section 7](#) derives its Möbius and product-slope consequences. [Section 8](#) gives the Q^3 transfer. The final sections record sharp obstructions, the literature boundary, the high- β ledger, and the exact computational certificate.

2 The physical zero-frequency interface

This section fixes notation for the physical object. It imports only the proved identities and norm bounds needed below; no estimate from a later section is used to redefine the packet.

2.1 Rows, targets, and actual coefficients

Fix $h \in \mathbb{Z} \setminus \{0\}$ and put $H = \text{rad } |h|$. On one opened dyadic packet,

$$Q = LD, \quad QJ \asymp_{\mathcal{D}} X, \quad R \leq T < U_0 \asymp_{\mathcal{D}} X, \quad (9)$$

where U_0 exceeds every target. A row is

$$\alpha = (\ell_\alpha, d_\alpha), \quad m_\alpha = \ell_\alpha d_\alpha \asymp_{\mathcal{D}} Q. \quad (10)$$

Here $\ell_\alpha \asymp L$ is prime, $d_\alpha \asymp D$ is squarefree, $(d_\alpha, H) = 1$, and the source separation makes the integer m_α determine its label. The two row coefficients are

$$\gamma_\alpha^{(i)} = \mu(d_\alpha)(\log \ell_\alpha)\omega_D^{(i)}(d_\alpha)\psi_L^{(i)}(\ell_\alpha/L)\zeta_\alpha^{(i)}, \quad i = 1, 2, \quad (11)$$

with fixed bounded smooth and residue factors. TPC-25 proves

$$\sum_\alpha |\gamma_\alpha^{(i)}| \ll_{\mathcal{D}} Q, \quad (12)$$

$$\sum_\alpha |\gamma_\alpha^{(i)}|^2 \ll_{\mathcal{D}} Q \log(2L). \quad (13)$$

For an orbit point j , let

$$N_\alpha(j) = m_\alpha j + h. \quad (14)$$

Every effective term has $(m_\alpha j, H) = 1$ and $N_\alpha(j) \asymp_{\mathcal{D}} X$. Hence every divisor $w \mid N_\alpha(j)$ satisfies

$$(w, m_\alpha j H) = 1. \quad (15)$$

The joint orbit support contains $O_{\mathcal{D}}(J)$ integers.

The bounded fixed-period and smooth factors are denoted by

$$\Xi_{\alpha, \gamma}(j) \mathcal{W}_{\alpha, \gamma}(j/J). \quad (16)$$

TPC-25 gives a finite residue separation and an absolutely convergent Mellin representation whose total projective mass is bounded by a fixed constant at zero frequency. We retain these factors rather than replace them by one.

2.2 Returning to the actual mask

The generic mask in TPC-21, and hence the actual mask studied here, is

$$\boxed{\mathbf{m}_{\text{act}}(\alpha, \gamma) = \mathbf{1}_{\ell_\alpha \neq \ell_\gamma} \mathbf{1}_{|m_\alpha - m_\gamma| > QX^{-\kappa_0}} \mathbf{1}_{(d_\alpha, d_\gamma) \leq G}, \quad G = X^{\kappa_0}.} \quad (17)$$

Here $\kappa_0 > 0$ is the fixed pruning parameter. Define

$$\mathfrak{A}_{\alpha, \gamma}^{\text{act}}(j) := \mathbf{m}_{\text{act}}(\alpha, \gamma) \Xi_{\alpha, \gamma}(j) \mathcal{W}_{\alpha, \gamma}(j/J). \quad (18)$$

It is uniformly bounded and vanishes on equal rows. All physical column theorems in this paper use (18). Statements valid only for an arbitrary entrywise bounded mask are explicitly labelled as abstract interfaces and are not used to infer the structured expansion in [section 5](#).

2.3 The exact matched polarization

Let

$$a_R(w) = -\mu(w) \log w \quad (19)$$

and let $b_R(w)$ be the matched residual divisor coefficient from TPC-32. Both vanish unless w is squarefree, and $b_R(w) = a_R(w)$ for $w > R$. Define

$$A_{m,Y}(j) := \sum_{\substack{w \leq Y \\ w|mj+h}} b_R(w), \quad (20)$$

$$C_m(j) := \sum_{\substack{T < w \leq U_0 \\ w|mj+h}} a_R(w). \quad (21)$$

Then

$$A_{m,U_0}(j) = A_{m,T}(j) + C_m(j). \quad (22)$$

Put

$$H_m(j) := \frac{A_{m,U_0}(j) + A_{m,T}(j)}{2} = A_{m,T}(j) + \frac{1}{2}C_m(j). \quad (23)$$

The three raw cutoff channels combine pointwise as

$$\boxed{\mathcal{K}_{\alpha,\gamma}^{\text{sh}}(j) = C_{m_\alpha}(j)H_{m_\gamma}(j) + H_{m_\alpha}(j)C_{m_\gamma}(j).} \quad (24)$$

Neither term may be discarded.

For an event $\mathcal{E}(\alpha, \gamma, j)$, write

$$\mathcal{S}_{\text{sh}}(\mathcal{E}) := \sum_{\alpha,\gamma} \gamma_\alpha^{(1)} \gamma_\gamma^{(2)} \sum_j \mathfrak{A}_{\alpha,\gamma}^{\text{act}}(j) \mathcal{K}_{\alpha,\gamma}^{\text{sh}}(j) \mathbf{1}_{\mathcal{E}(\alpha,\gamma,j)}, \quad (25)$$

$$\mathcal{S}_{\text{sh}}^{\text{all}} := \mathcal{S}_{\text{sh}}(\text{all}). \quad (26)$$

Its natural absolute scale is

$$N_0 = JQ^2 \asymp XQ. \quad (27)$$

If

$$c_{\alpha,\gamma}(j) = (N_\alpha(j), N_\gamma(j)), \quad (28)$$

then TPC-30 and TPC-32 prove [13, 14], for $1 \leq C \ll_{\mathscr{D}} Q$ in the range of the inherited theorem,

$$\mathcal{S}_{\text{sh}}^{\text{all}} = \widehat{A}_{C,q}(0) + \mathcal{S}_{\text{sh}}(c_{\alpha,\gamma}(j) > C), \quad (29)$$

$$|\mathcal{S}_{\text{sh}}(c_{\alpha,\gamma}(j) > C)| \ll_{\varepsilon,h,\mathscr{D}} X^\varepsilon \left(Q^2 + \frac{XQ}{C} \right). \quad (30)$$

The clean affine expansion below is performed on $\mathcal{S}_{\text{sh}}^{\text{all}}$. Whenever $J \ll_{\mathscr{D}} Q$, in particular at the high- β endpoint studied here, choosing $C \asymp J$ restores the desired small-content zero coefficient with an $O(X^\varepsilon Q^2)$ error. This order avoids introducing the content-projector Möbius factor into the affine sign product.

Finally, let $\mathcal{D}^{\text{dr}}(\mathcal{E})$ denote, with the sign convention of TPC-32, the contribution of the two additional drift legs under an event $\mathcal{E}$, and put $\mathcal{D}^{\text{dr}} = \mathcal{D}^{\text{dr}}(\text{all})$. Their already proved uniform bound is

$$|\mathcal{D}^{\text{dr}}| \ll_{\varepsilon,h,\mathscr{D}} X^\varepsilon \frac{XQ}{L}. \quad (31)$$

They will be restored only after the raw matched shell has been estimated.

3 Integral normal form on a fixed-determinant line

We isolate the arithmetic geometry used by every affine column. Throughout this section,

$$h \in \mathbb{Z} \setminus \{0\}, \quad a, s \in \mathbb{N}, \quad (a, s) = 1, \quad (as, h) = 1. \quad (32)$$

3.1 The $SL_2(\mathbb{Z})$ parametrization

Choose $x, y \in \mathbb{Z}$ with

$$sx - ay = 1. \quad (33)$$

Set

$$V = \begin{pmatrix} x & a \\ y & s \end{pmatrix} \in SL_2(\mathbb{Z}). \quad (34)$$

Theorem 3.1 (Fixed-determinant normal form). *Every integral solution of*

$$sU - aD = h \quad (35)$$

has a unique representation

$$\boxed{U_t = hx + at, \quad D_t = hy + st, \quad t \in \mathbb{Z}.} \quad (36)$$

The inverse coordinates are

$$\boxed{\begin{pmatrix} h \\ t \end{pmatrix} = \begin{pmatrix} s & -a \\ -y & x \end{pmatrix} \begin{pmatrix} U \\ D \end{pmatrix}.} \quad (37)$$

For every t , with gcds understood positively,

$$\boxed{(D_t, U_t) = (D_t, h) = (U_t, h) = (h, t).} \quad (38)$$

Moreover,

$$(D_t, s) = 1, \quad (U_t, a) = 1. \quad (39)$$

Proof. The determinant of V is one. Multiplying $V(h, t)^T$ gives (36), and $sU_t - aD_t = (sx - ay)h = h$. Conversely, the inverse matrix in (37) is integral. Its first coordinate is $sU - aD = h$, so every solution has one and only one integral second coordinate t .

An integral unimodular transformation preserves the ideal generated by the two coordinates. Hence $(D_t, U_t) = (h, t)$. Also

$$(D_t, h) = (st, h) = (t, h), \quad (U_t, h) = (at, h) = (t, h),$$

because $(as, h) = 1$. This proves (38). Finally, (33) implies $(y, s) = (x, a) = 1$. Therefore

$$(D_t, s) = (hy, s) = 1, \quad (U_t, a) = (hx, a) = 1,$$

again using $(as, h) = 1$. □

Every other Bézout pair is

$$x' = x + ak, \quad y' = y + sk, \quad k \in \mathbb{Z}. \quad (40)$$

For a fixed physical solution, changing from (x, y) to (x', y') changes its affine coordinate from t to $t - hk$. Thus the residue class $t \pmod{h}$, and in particular (t, h) , is intrinsic even though the integer origin is not.

Corollary 3.2 (Primitive layer). *The following conditions are equivalent:*

$$(D_t, h) = 1, \quad (U_t, h) = 1, \quad (D_t, U_t) = 1, \quad (t, h) = 1. \quad (41)$$

Proof. All four gcds are equal by (38). □

3.2 Möbius layers and the quadratic discriminant

Let

$$g = (h, t) = (D_t, U_t), \quad D_t = gD'_t, \quad U_t = gU'_t. \quad (42)$$

Then $(D'_t, U'_t) = 1$.

Proposition 3.3 (Exact Möbius gcd layer). *Whenever $D_t, U_t > 0$,*

$$\boxed{\mu(D_t)\mu(U_t) = \mu^2(g)\mathbf{1}_{(g, D'_t U'_t)=1}\mu(D'_t U'_t)}. \quad (43)$$

In particular, on the primitive layer $(t, h) = 1$,

$$\boxed{\mu(D_t)\mu(U_t) = \mu(D_t U_t)}. \quad (44)$$

Proof. If g is not squarefree or shares a prime with $D'_t U'_t$, both sides of (43) vanish. Otherwise g, D'_t, U'_t are pairwise coprime and

$$\mu(gD'_t)\mu(gU'_t) = \mu(g)^2\mu(D'_t)\mu(U'_t) = \mu^2(g)\mu(D'_t U'_t).$$

The primitive statement follows from $g = 1$. □

The fused polynomial is

$$F_{a,s,h}(t) := D_t U_t = ast^2 + h(sx + ay)t + h^2 xy. \quad (45)$$

Proposition 3.4 (Square discriminant). *The discriminant of $F_{a,s,h}$ is exactly*

$$\boxed{\text{disc } F_{a,s,h} = h^2}. \quad (46)$$

Proof. By (33),

$$\begin{aligned} \text{disc } F_{a,s,h} &= h^2((sx + ay)^2 - 4asxy) \\ &= h^2(sx - ay)^2 = h^2. \end{aligned}$$

□

Thus the two affine signs form a reducible quadratic Möbius sequence with a fixed square discriminant. The identity is structural, not a cancellation theorem: the case $a = s = h = 1$ contains the open sequence $\mu(t)\mu(t+1)$, as recorded in [section 9.1](#).

3.3 The reduced physical origin

For column disintegration it is often more convenient to choose the unique integer D_0 with

$$1 \leq D_0 \leq s, \quad aD_0 + h \equiv 0 \pmod{s}, \quad (47)$$

and put

$$U_0 = \frac{aD_0 + h}{s}. \quad (48)$$

Then all solutions are

$$D_r = D_0 + sr, \quad U_r = U_0 + ar, \quad (49)$$

with

$$|D_0| \leq s, \quad |U_0| \leq a + |h|. \quad (50)$$

This origin differs from the canonical one by an integral translation. Accordingly, $(D_r, h) = 1$ excludes one translated residue class of $r \pmod{p}$ for every $p \mid h$; it should not in general be replaced by the literal condition $(r, h) = 1$.

4 A balanced moving-slope squarefree-energy theorem

The normal form separates a support question from a sign question. We now show that the squarefree support has the expected positive density in a genuine growing-slope range. The proof is an elementary squarefree sieve tailored to the Bézout coordinates. The origin must be balanced: an arbitrary transformation $(x, y) \mapsto (x + ak, y + sk)$ can make the heights of both forms arbitrarily large while a, s , and the length of the parameter interval remain fixed. For general background on elementary squarefree sieves, see Iwaniec and Kowalski [3].

4.1 Balanced coordinates and the Euler product

Assume (32). If $a \geq 2$, choose the unique integer x such that

$$sx \equiv 1 \pmod{a}, \quad -\frac{a}{2} < x \leq \frac{a}{2}, \quad (51)$$

and put $y = (sx - 1)/a$. If $a = 1$, take $x = 0, y = -1$. Then

$$sx - ay = 1, \quad |x| \leq \frac{a}{2}, \quad |y| \leq \frac{s}{2} + 1. \quad (52)$$

Use the forms D_t, U_t from (36) and put

$$K = \max(a, s). \quad (53)$$

For $T \geq 4(|h| + 1)$ and $T < t \leq 2T$, both forms are positive and

$$\max(D_t, U_t) \ll_h KT. \quad (54)$$

Define

$$E_{a,s,h}(T) := \sum_{\substack{T < t \leq 2T \\ (t,h)=1}} \mu^2(D_t) \mu^2(U_t) \quad (55)$$

and

$$\boxed{\mathfrak{S}_{\text{sf}}(a, s; h) := \prod_{p|h} \left(1 - \frac{1}{p}\right) \prod_{\substack{p|h \\ p \nmid as}} \left(1 - \frac{1}{p^2}\right) \prod_{p \nmid ash} \left(1 - \frac{2}{p^2}\right)}. \quad (56)$$

Theorem 4.1 (Uniform squarefree energy). *For fixed $h \neq 0$, uniformly over all positive a, s, T satisfying (32) and $T \geq 4(|h| + 1)$,*

$$\boxed{E_{a,s,h}(T) = \mathfrak{S}_{\text{sf}}(a, s; h)T + O_h\left(\frac{T}{\log(2T)} + \sqrt{KT}\right)}. \quad (57)$$

Furthermore,

$$\mathfrak{S}_{\text{sf}}(a, s; h) \geq c_h > 0 \quad (58)$$

uniformly in a, s . Consequently, for every fixed $\eta > 0$,

$$K \leq T^{1-\eta} \implies E_{a,s,h}(T) = \mathfrak{S}_{\text{sf}}(a, s; h)T + o_{h,\eta}(T) \quad (59)$$

uniformly in that range.

4.2 Local root counts

Lemma 4.2 (The three local factors). *The local density of the conditions in (55) is:*

- (i) $1 - 1/p$ if $p \mid h$;
- (ii) $1 - 1/p^2$ if $p \nmid h$ and $p \mid as$;
- (iii) $1 - 2/p^2$ if $p \nmid ash$.

Proof. If $p \mid h$, then $p \nmid as$. The primitive condition excludes $t \equiv 0 \pmod{p}$, and

$$D_t \equiv st \pmod{p}, \quad U_t \equiv at \pmod{p}.$$

Thus neither form is divisible by p on the retained classes, giving $1 - 1/p$.

Suppose $p \nmid h$ and $p \mid s$. Then $p \nmid a$, and $sx - ay = 1$ makes $y \not\equiv 0 \pmod{p}$. Hence $D_t \equiv hy \not\equiv 0 \pmod{p}$, while U_t has exactly one root modulo p^2 . The case $p \mid a$ is symmetric. This proves (ii).

Finally suppose $p \nmid ash$. Each affine form has exactly one root modulo p^2 . The roots are distinct: a common root would imply $p^2 \mid sU_t - aD_t = h$, which is impossible. Thus precisely two classes modulo p^2 are removed. The argument includes $p = 2$. $\square$

4.3 Small primes and the large-square tail

Proof of theorem 4.1. For $z \geq 2$, put

$$M_z = \text{rad } |h| \prod_{\substack{p \leq z \\ p \nmid h}} p^2. \quad (60)$$

Let $N_z(T)$ count $T < t \leq 2T$ for which $(t, h) = 1$ and neither form is divisible by p^2 for any $p \leq z$, $p \nmid h$. The conditions are periodic modulo M_z . By the Chinese remainder theorem and lemma 4.2,

$$N_z(T) = \mathfrak{S}_z(a, s; h)T + O(M_z), \quad (61)$$

where $\mathfrak{S}_z$ is the product in (56) truncated at primes at most z , together with all fixed factors $p \mid h$. Absolute convergence gives, uniformly in a, s ,

$$|\mathfrak{S}_z(a, s; h) - \mathfrak{S}_{\text{sf}}(a, s; h)| \ll_h \sum_{p > z} \frac{1}{p^2} \ll_h \frac{1}{z}. \quad (62)$$

It remains to remove integers for which a prime $p > z$ has $p^2 \mid D_t U_t$. If $p \mid s$, the congruence $p^2 \mid D_t$ has no solution; otherwise it occupies at most one class modulo p^2 . The analogous statement holds for U_t . By (54), a relevant prime satisfies $p \ll_h \sqrt{KT}$. Therefore

$$\begin{aligned} |N_z(T) - E_{a,s,h}(T)| &\ll_h \sum_{z < p \ll_h \sqrt{KT}} \left(\frac{T}{p^2} + 1 \right) \\ &\ll_h \frac{T}{z} + \sqrt{KT}. \end{aligned} \quad (63)$$

For all sufficiently large T , Chebyshev's estimate $\sum_{p \leq z} \log p \ll z$ permits the choice $z = c \log(2T)$, with a sufficiently small absolute $c > 0$, for which $M_z \ll_h T^{1/2}$; the remaining bounded range is absorbed by the implied constant. Combining (61)–(63), and absorbing the harmless $T^{1/2}$ term, proves (57).

For positivity, replacing a factor $1 - 2/p^2$ by $1 - 1/p^2$ can only increase the product. Hence

$$\mathfrak{S}_{\text{sf}}(a, s; h) \geq \prod_{p|h} \left(1 - \frac{1}{p}\right) \prod_{p \nmid h} \left(1 - \frac{2}{p^2}\right) =: c_h > 0. \quad (64)$$

Finally, if $K \leq T^{1-\eta}$, the second error in (57) is at most $T^{1-\eta/2}$, proving (59). $\square$

Remark 4.3 (What the theorem does not give). The summand in (55) is nonnegative. The theorem proves that squarefree values persist uniformly in the stated moving-slope window; it gives no cancellation in $\sum \mu(D_t)\mu(U_t)$. In the physical high- β columns the slope can be much larger than the affine-fiber length, so even the asymptotic range $K \leq T^{1-\eta}$ need not apply.

Remark 4.4 (Why the dyadic origin matters). For a general interval $T_0 < t \leq T_0 + Y$, the same tail argument naturally contains $\sqrt{K(|T_0| + Y)}$, not merely $\sqrt{KY}$. The balanced dyadic formulation is therefore a genuine hypothesis rather than a cosmetic normalization.

5 Structured-mask disintegration of the physical columns

We now prove the physical reduction announced in (2). The result uses the actual mask (17). It is false as a statement about every entrywise bounded mask.

5.1 An exact projection of the row-gcd mask

For $G \geq 1$, define

$$\lambda_G(v) := \sum_{\substack{g|v \\ g \leq G}} \mu(v/g). \quad (65)$$

Lemma 5.1 (Exact row-gcd projector). *For every $d, e \in \mathbb{N}$,*

$$\boxed{\mathbf{1}_{(d,e) \leq G} = \sum_{\substack{v|d \\ v|e}} \lambda_G(v).} \quad (66)$$

If e is squarefree, then

$$\boxed{\sum_{v|e} |\lambda_G(v)| \leq 3^{\omega(e)} \ll_{\varepsilon} e^{\varepsilon}.} \quad (67)$$

Proof. Put $n = (d, e)$. Reversing the two divisor sums gives

$$\begin{aligned} \sum_{v|n} \lambda_G(v) &= \sum_{\substack{g|n \\ g \leq G}} \sum_{w|n/g} \mu(w) \\ &= \mathbf{1}_{n \leq G}. \end{aligned}$$

This proves (66). If e is squarefree, then

$$\sum_{v|e} |\lambda_G(v)| \leq \sum_{v|e} \sum_{g|v} |\mu(v/g)| = \sum_{gw|e} 1 = 3^{\omega(e)}.$$

The final bound is the standard divisor estimate. $\square$

The key point is the order of use. After the opposite row is fixed, v ranges only over divisors of its squarefree coordinate e , so the total coefficient mass is $X^{o(1)}$. Summing (66) without fixing a column would not by itself prove a small operator norm for the full mask.

5.2 Reflecting one ultra leg

Fix a right row

$$\gamma = (\ell', e), \quad n = m_\gamma = \ell' e, \quad (68)$$

and consider the left-ultra column

$$B_\gamma^L := \sum_\alpha \gamma_\alpha^{(1)} \sum_j \mathfrak{A}_{\alpha, \gamma}^{\text{act}}(j) C_{m_\alpha}(j) H_{m_\gamma}(j). \quad (69)$$

Open the ultra increment. For a nonzero term with $m_\alpha = \ell d$, let u be its ultra divisor and put

$$N_\alpha(j) = \ell d j + h = u s. \quad (70)$$

Then

$$T < u \leq U_0, \quad 1 \leq s < \frac{N_\alpha(j)}{T} \ll_{\mathscr{D}} \frac{X}{T}. \quad (71)$$

The variable s is not required to be squarefree. Target primitivity (15) gives

$$(s, \ell d j H) = (u, \ell d j H) = 1. \quad (72)$$

It does not imply $(s, u) = 1$.

Fix ℓ, j, s . Since $(s, \ell j) = 1$, there is one residue $d_0 \pmod{s}$ satisfying

$$\ell j d_0 + h \equiv 0 \pmod{s}. \quad (73)$$

Writing $d = d_0 + s\tau$ gives

$$u = u_0 + \ell j \tau, \quad u_0 = \frac{\ell j d_0 + h}{s}, \quad s u_0 - \ell j d_0 = h. \quad (74)$$

The physical dyadic restriction $d \asymp D$ permits at most

$$O\left(1 + \frac{D}{s}\right) \quad (75)$$

values of τ .

5.3 The full actual-mask column theorem

The row-gcd projector introduces the divisor v . On a nonempty term, $v \mid d, e$, and (72) implies $(v, s) = 1$. Write

$$d = v \tilde{d}. \quad (76)$$

Since $(s, \ell j v) = 1$, there is a unique $\tilde{d}_0 \pmod{s}$ such that

$$\ell j v \tilde{d}_0 + h \equiv 0 \pmod{s}. \quad (77)$$

Put

$$\tilde{u}_0 = \frac{\ell j v \tilde{d}_0 + h}{s}. \quad (78)$$

Theorem 5.2 (Actual-mask affine-column disintegration). *Fix the right row $\gamma = (\ell', e)$ in (69). After opening the left ultra leg, the actual mask decomposes that column into the variables ℓ, j, s, v , with $v \mid e$, and at most two intervals I on each such layer. The inner arithmetic sequence has the exact form*

$$\boxed{\sum_{r \in I} \mu(\tilde{d}_0 + sr) \mu(\tilde{u}_0 + \ell j v r) \mathbf{1}_{(v, (\tilde{d}_0 + sr)(\tilde{u}_0 + \ell j v r))=1} \rho(r) W(r)}, \quad (79)$$

where

$$s\tilde{u}_0 - \ell j v \tilde{d}_0 = h, \quad (80)$$

$$|I| \ll_{\mathcal{D}} 1 + \frac{D}{sv}. \quad (81)$$

Here ρ is a bounded fixed-period factor. The weight $W(r)$ includes the varying factor $\log u_r$, the row smooth factors, and their Mellin/BV separations inherited from TPC-25; its seminorm cost is therefore at most logarithmic in addition to the inherited soft losses. The coefficient outside (79) contains $-\mu(v)\lambda_G(v)(\log \ell)$ and the remaining fixed physical factors. Uniformly in the fixed column,

$$\sum_{v|e} |\lambda_G(v)| \ll_{\varepsilon} X^{\varepsilon}. \quad (82)$$

The local coprimality indicator in (79) itself has a divisor expansion of total coefficient mass at most $2^{\omega(v)} \ll_{\varepsilon} X^{\varepsilon}$.

Proof. Insert (66) into the last factor of (17). Because e is fixed and squarefree, (67) proves (82). For one divisor $v | e$, write $d = v\tilde{d}$ and use (77)–(78). Every solution is uniquely

$$\tilde{d}_r = \tilde{d}_0 + sr, \quad u_r = \tilde{u}_0 + \ell j v r, \quad (83)$$

and these forms satisfy (80). The interval $d = v\tilde{d} \asymp D$ has length $O(D/v)$; sampling it with step s proves (81).

For fixed ℓ, ℓ', e, v , the same-source condition is constant. The near-diagonal deletion is

$$|\ell v \tilde{d} - \ell' e| > QX^{-\kappa_0}.$$

Its complement is one interval on the $\tilde{d}$ -line, so the retained set is the union of at most two intervals. The fixed residue factor is periodic with a modulus depending only on h , and the smooth factors have the Mellin/BV separations already recorded in TPC-25. These give ρ and W without an additional polynomial projective cost, apart from the logarithmic and soft losses recorded above.

The original row is squarefree, so v and $\tilde{d}_r$ are squarefree and coprime. The ultra coefficient is supported on squarefree u_r , and target primitivity makes $(v, u_r) = 1$. Therefore

$$\mu(d)(-\mu(u_r) \log u_r) = -\mu(v)\mu(\tilde{d}_r)\mu(u_r) \log u_r \quad (84)$$

on the physical support. Restoring the coprimality by the displayed indicator yields (79). Finally,

$$\mathbf{1}_{(v,n)=1} = \sum_{c|(v,n)} \mu(c)$$

has coefficient mass $2^{\omega(v)}$, which is $X^{o(1)}$ because $v | e \ll X^{O(1)}$. $\square$

Corollary 5.3 (Both polarizations). *The right-ultra column obtained by fixing α and opening C_{m_γ} has the symmetric disintegration (79), with the two row labels exchanged. Thus both terms in (24) are covered.*

Remark 5.4 (Column complexity versus matrix complexity). The theorem fixes the opposite row before expanding the gcd mask. It does not give an $X^{o(1)}$ projective decomposition of the full matrix $(\mathbf{m}_{\text{act}}(\alpha, \gamma))$. In particular, it cannot be substituted into an abstract Schur-multiplier argument without retaining the outer column energy.

Remark 5.5 (Fixed determinant, growing slopes). The two slopes in (79) are s and $a = \ell j v$. Both are coprime to h and to one another, and their determinant is the fixed integer h . They nevertheless grow with X , and the interval may contain only $O(1)$ points when $sv > D$. A theorem for one fixed affine pair cannot simply be summed over these layers.

6 Fixed-determinant projective Plancherel

We pass to a finite model in which the two-parameter affine coefficient family is complete. The terminology “projective Plancherel” is descriptive: the proof is finite Fourier orthogonality in the coordinates $b = a/s$ and $c = h/s$. Thus the second moment averages both the projective ratio b and the nonzero affine offset c ; it gives no almost-every- b statement by itself. The useful feature here is its specialization to one nonzero determinant fiber and the explicit defect left by homogeneous dilations.

Let q be prime, let $h \in \mathbb{F}_q^\times$, and let $f, g : \mathbb{F}_q \rightarrow \mathbb{C}$. We use the unnormalized conventions

$$\bar{f} = \frac{1}{q} \sum_{x \in \mathbb{F}_q} f(x), \quad f^\circ = f - \bar{f}, \quad (85)$$

$$\|f\|_2^2 = \sum_{x \in \mathbb{F}_q} |f(x)|^2, \quad \hat{f}(r) = \sum_{x \in \mathbb{F}_q} f(x) e_q(-rx), \quad (86)$$

where $e_q(z) = \exp(2\pi iz/q)$. For $a, s \in \mathbb{F}_q^\times$, define

$$\mathcal{C}_{a,s}^{(h)}(f, g) := \sum_{D \in \mathbb{F}_q} f(D) g(s^{-1}(aD + h)). \quad (87)$$

This is the correlation along the affine line $sU - aD = h$.

Theorem 6.1 (Exact determinant-fiber second moment). *With the preceding conventions,*

$$\boxed{\sum_{a,s \in \mathbb{F}_q^\times} \left| \mathcal{C}_{a,s}^{(h)}(f, g) - q\bar{f}\bar{g} \right|^2} = q\|f^\circ\|_2^2 \|g^\circ\|_2^2 - \sum_{b \in \mathbb{F}_q^\times} \left| \sum_{D \in \mathbb{F}_q} f^\circ(D) g^\circ(bD) \right|^2. \quad (88)$$

In particular,

$$\boxed{\sum_{a,s \in \mathbb{F}_q^\times} \left| \mathcal{C}_{a,s}^{(h)}(f, g) - q\bar{f}\bar{g} \right|^2} \leq q\|f^\circ\|_2^2 \|g^\circ\|_2^2. \quad (89)$$

Proof. Put

$$b = a/s, \quad c = h/s. \quad (90)$$

Because $h \neq 0$, the map $(a, s) \mapsto (b, c)$ is a bijection of $(\mathbb{F}_q^\times)^2$: its inverse is $s = h/c$, $a = bs$. Centering both functions gives

$$\mathcal{C}_{a,s}^{(h)}(f, g) - q\bar{f}\bar{g} = R_{b,c}, \quad R_{b,c} := \sum_D f^\circ(D) g^\circ(bD + c). \quad (91)$$

The mixed term containing g° vanishes because an affine permutation preserves its zero sum, while the other vanishes because $\sum_D f^\circ(D) = 0$.

Temporarily include $c = 0$. Fourier transformation in c gives

$$\hat{R}_b(r) := \sum_{c \in \mathbb{F}_q} R_{b,c} e_q(-rc) = \hat{f}^\circ(-br) \hat{g}^\circ(r). \quad (92)$$

Parseval and $\hat{f}^\circ(0) = \hat{g}^\circ(0) = 0$ imply

$$\sum_{b \neq 0} \sum_{c \in \mathbb{F}_q} |R_{b,c}|^2 = \frac{1}{q} \sum_{r \neq 0} |\hat{g}^\circ(r)|^2 \sum_{b \neq 0} |\hat{f}^\circ(-br)|^2$$

$$\begin{aligned}
&= \frac{1}{q} \left(\sum_{\rho \neq 0} |\hat{f}^\circ(\rho)|^2 \right) \left(\sum_{r \neq 0} |\hat{g}^\circ(r)|^2 \right) \\
&= q \|f^\circ\|_2^2 \|g^\circ\|_2^2.
\end{aligned} \tag{93}$$

The determinant- h family has $c \neq 0$. Subtracting its omitted $c = 0$ energy,

$$\sum_{b \neq 0} |R_{b,0}|^2 = \sum_{b \neq 0} \left| \sum_D f^\circ(D) g^\circ(bD) \right|^2,$$

proves (88). Dropping this nonnegative term proves (89). $\square$

Remark 6.2 (The dilation-alignment defect). The subtracted term is the complete energy of the homogeneous lines $U = bD$. Thus the identity can be strictly smaller than its upper bound when f° and a multiplicative dilation of g° align. No randomness assumption is made.

Corollary 6.3 (Exceptional determinant-fiber pairs). *For every $\Lambda > 0$,*

$$\begin{aligned}
&\# \left\{ (a, s) \in (\mathbb{F}_q^\times)^2 : \left| \mathcal{C}_{a,s}^{(h)}(f, g) - q \bar{f} \bar{g} \right| > \Lambda \right\} \\
&\leq \frac{q \|f^\circ\|_2^2 \|g^\circ\|_2^2}{\Lambda^2}.
\end{aligned} \tag{94}$$

Proof. Apply Chebyshev's inequality to (89). $\square$

The corollary is an average over the two-dimensional coefficient family (a, s) , equivalently over (b, c) . It gives no pointwise cancellation for one prescribed physical pair (a, s) , nor an almost-every- b conclusion, unless an additional argument controls the remaining coordinate.

7 Finite Möbius slopes and prime-product lifts

We derive two consequences of the exact second moment. The first is an unconditional almost-all-pair theorem for a finite Möbius model. The second verifies that a complete product slope $a = \ell j$ does not lose projective coverage merely because ℓ is restricted.

7.1 A finite Möbius model

For a prime q , define $\mu_q : \mathbb{F}_q \rightarrow \mathbb{R}$ by

$$\mu_q(0) = 0, \quad \mu_q(n) = \mu(n), \quad 1 \leq n < q, \tag{95}$$

using the standard representatives. Let

$$M(x) = \sum_{n \leq x} \mu(n). \tag{96}$$

Then

$$\overline{\mu_q} = \frac{M(q-1)}{q}, \quad \|\mu_q^\circ\|_2^2 \leq q. \tag{97}$$

Theorem 7.1 (Almost every finite determinant-fiber Möbius pair). *Fix $0 < \eta < 1/2$. For every prime q and every $h \in \mathbb{F}_q^\times$, outside a set $\mathcal{E}_{q,h,\eta} \subset (\mathbb{F}_q^\times)^2$ satisfying*

$$|\mathcal{E}_{q,h,\eta}| \leq q^{2-2\eta}, \tag{98}$$

one has

$$\mathcal{C}_{a,s}^{(h)}(\mu_q, \mu_q) = \frac{M(q-1)^2}{q} + O(q^{1/2+\eta}). \quad (99)$$

Consequently,

$$\mathcal{C}_{a,s}^{(h)}(\mu_q, \mu_q) = o(q) \quad (100)$$

for a proportion $1 - o(1)$ of coefficient pairs (a, s) as $q \rightarrow \infty$.

Proof. Apply [corollary 6.3](#) with $f = g = \mu_q$ and $\Lambda = q^{1/2+\eta}$. By [\(97\)](#), the total variance is at most q^3 , so the exceptional count is at most $q^3/q^{1+2\eta} = q^{2-2\eta}$. The centered main term is

$$q\overline{\mu_q}^2 = \frac{M(q-1)^2}{q},$$

which proves [\(99\)](#). The prime number theorem is equivalent to $M(x) = o(x)$, so this main term is $o(q)$; the error is also $o(q)$ because $\eta < 1/2$. $\square$

Remark 7.2 (Model boundary). The map $n \mapsto \mu_q(n)$ is not a periodic multiplicative function on the integers; it is the finite vector obtained by truncating the ordinary Möbius function before q . The theorem is therefore a modular model for determinant-fiber coefficient averaging, not a theorem about one integer affine fiber.

7.2 Complete and incomplete product slopes

Let $\mathcal{L} \subset \mathbb{F}_q^\times$ be arbitrary. For a fixed $\ell \in \mathcal{L}$, multiplication by ℓ permutes $\mathbb{F}_q^\times$.

Proposition 7.3 (Exact product-slope lift). *For every $f, g : \mathbb{F}_q \rightarrow \mathbb{C}$,*

$$\begin{aligned} & \sum_{\ell \in \mathcal{L}} \sum_{j, s \in \mathbb{F}_q^\times} \left| \mathcal{C}_{\ell j, s}^{(h)}(f, g) - q\bar{f}\bar{g} \right|^2 \\ &= |\mathcal{L}| \sum_{a, s \in \mathbb{F}_q^\times} \left| \mathcal{C}_{a, s}^{(h)}(f, g) - q\bar{f}\bar{g} \right|^2. \end{aligned} \quad (101)$$

In particular, restricting $\mathcal{L}$ to prime residue classes causes no coverage loss when j runs through a complete nonzero residue system.

Proof. For each fixed $\ell \neq 0$, the substitution $a = \ell j$ is a bijection of $\mathbb{F}_q^\times$. Apply it inside the j -sum and then sum over ℓ . $\square$

For an incomplete integer set $\mathcal{J}$, let

$$\nu_{\mathcal{J}}(r) = \#\{j \in \mathcal{J} : j \equiv r \pmod{q}\}, \quad \nu_{\mathcal{J}}^* = \max_{r \in \mathbb{F}_q^\times} \nu_{\mathcal{J}}(r), \quad (102)$$

and define $\nu_{\mathcal{S}}^*$ similarly. We restrict to elements nonzero modulo q .

Corollary 7.4 (Incomplete product-slope energy). *For finite integer sets $\mathcal{J}, \mathcal{S}$ with no multiples of q ,*

$$\begin{aligned} & \sum_{\ell \in \mathcal{L}} \sum_{j \in \mathcal{J}} \sum_{s \in \mathcal{S}} \left| \mathcal{C}_{\ell j, s}^{(h)}(f, g) - q\bar{f}\bar{g} \right|^2 \\ & \leq |\mathcal{L}| \nu_{\mathcal{J}}^* \nu_{\mathcal{S}}^* q \|f^\circ\|_2^2 \|g^\circ\|_2^2. \end{aligned} \quad (103)$$

If $\mathcal{J}, \mathcal{S}$ are intervals of respective lengths J, S , then

$$\nu_{\mathcal{J}}^* \leq J/q + 1, \quad \nu_{\mathcal{S}}^* \leq S/q + 1. \quad (104)$$

After division by $|\mathcal{L}|JS$, the rms loss relative to complete residue averaging is at most

$$\boxed{\left(1 + \frac{q}{J}\right)^{1/2} \left(1 + \frac{q}{S}\right)^{1/2}}. \quad (105)$$

Proof. Group j, s by their nonzero residues. Each pair is repeated at most $\nu_{\mathcal{J}}^* \nu_{\mathcal{S}}^*$ times. Apply [proposition 7.3](#) and then (89). The interval bounds are elementary. Dividing the resulting mean square by the complete-residue mean-square normalization introduces an additional factor $(q-1)/q \leq 1$; discarding it gives (105). $\square$

The exact lift is useful only after the orbit variable covers residue classes economically. When $q \gg J$, the first factor in (105) is large; this is the coverage side of the single-modulus obstruction proved later.

8 The physical Q^3 column-energy gate

The affine disintegration identifies the correct inner sequences. We now state exactly how much collective cancellation is needed from them. The outer row coefficient must not be included twice in the column energy.

8.1 The two polarized column energies

For a fixed right row γ , define the left-ultra column

$$B_{\gamma}^L := \sum_{\alpha} \gamma_{\alpha}^{(1)} \sum_j \mathfrak{A}_{\alpha, \gamma}^{\text{act}}(j) C_{m_{\alpha}}(j) H_{m_{\gamma}}(j), \quad (106)$$

and its energy

$$E_L := \sum_{\gamma} |B_{\gamma}^L|^2. \quad (107)$$

The coefficient $\gamma_{\gamma}^{(2)}$ is outside B_{γ}^L . Symmetrically, for a fixed left row α , put

$$B_{\alpha}^R := \sum_{\gamma} \gamma_{\gamma}^{(2)} \sum_j \mathfrak{A}_{\alpha, \gamma}^{\text{act}}(j) H_{m_{\alpha}}(j) C_{m_{\gamma}}(j), \quad (108)$$

$$E_R := \sum_{\alpha} |B_{\alpha}^R|^2. \quad (109)$$

Here $\gamma_{\alpha}^{(1)}$ is outside B_{α}^R . Define the two scalar polarizations

$$\mathcal{S}_L = \sum_{\gamma} \gamma_{\gamma}^{(2)} B_{\gamma}^L, \quad \mathcal{S}_R = \sum_{\alpha} \gamma_{\alpha}^{(1)} B_{\alpha}^R. \quad (110)$$

The exact identity (24) gives

$$\mathcal{S}_{\text{sh}}^{\text{all}} = \mathcal{S}_L + \mathcal{S}_R. \quad (111)$$

Theorem 8.1 (Matched polarized column-energy transfer). *On the physical packet of [section 2](#),*

$$\boxed{|\mathcal{S}_{\text{sh}}^{\text{all}}| \ll_{\mathscr{D}} \sqrt{Q \log(2L)} \left(\sqrt{E_L} + \sqrt{E_R} \right)}. \quad (112)$$

Consequently, the additional estimate

$$\boxed{E_L + E_R \ll_{\varepsilon, h, \mathcal{D}} X^\varepsilon Q^3} \quad (113)$$

for every fixed $\varepsilon > 0$ implies

$$\boxed{|\mathcal{S}_{\text{sh}}^{\text{all}}| \ll_{\varepsilon, h, \mathcal{D}} X^\varepsilon Q^2.} \quad (114)$$

Proof. By (110), Cauchy–Schwarz, and the inherited row energy (13),

$$\begin{aligned} |\mathcal{S}_L| &\leq \left(\sum_{\gamma} |\gamma_{\gamma}^{(2)}|^2 \right)^{1/2} \left(\sum_{\gamma} |B_{\gamma}^L|^2 \right)^{1/2} \\ &\ll_{\mathcal{D}} \sqrt{Q \log(2L)} \sqrt{E_L}, \end{aligned} \quad (115)$$

and similarly

$$|\mathcal{S}_R| \ll_{\mathcal{D}} \sqrt{Q \log(2L)} \sqrt{E_R}. \quad (116)$$

Use (111) and the triangle inequality to obtain (112). If (113) holds, then

$$\sqrt{E_L} + \sqrt{E_R} \leq \sqrt{2(E_L + E_R)} \ll X^{\varepsilon/2} Q^{3/2}.$$

Applying the input with a smaller soft exponent and absorbing $\log(2L)$ proves (114). $\square$

Remark 8.2 (Sufficient, not equivalent). The estimate (113) controls the two polarizations separately. It is a clean sufficient condition for the matched scalar shell, but cancellation between $\mathcal{S}_L$ and $\mathcal{S}_R$ could make the scalar sum smaller even when one energy is large. No necessity or equivalence is asserted.

8.2 Splicing back to the distinguished zero coefficient

Assume $J \ll_{\mathcal{D}} Q$, as holds at the high- β endpoint in section 10. Then $C = \lfloor J \rfloor$ lies in the inherited range of (29). Since $X \asymp QJ$,

$$Q^2 + \frac{XQ}{J} \asymp Q^2. \quad (117)$$

Corollary 8.3 (Conditional zero-frequency closure). *Suppose $J \ll_{\mathcal{D}} Q$. If (113) holds, then the small-content distinguished coefficient from TPC-32 satisfies*

$$\boxed{|\hat{A}_{\lfloor J \rfloor, q}(0)| \ll_{\varepsilon, h, \mathcal{D}} X^\varepsilon Q^2.} \quad (118)$$

With the inherited sign convention, define the complete calibrated cutoff difference by

$$\mathcal{S}_{\text{cal}}^{\text{full}} := \mathcal{S}_{\text{sh}}^{\text{all}} + \mathcal{D}^{\text{dr}}(\text{all}). \quad (119)$$

It satisfies

$$\boxed{|\mathcal{S}_{\text{cal}}^{\text{full}}| \ll_{\varepsilon, h, \mathcal{D}} X^\varepsilon \left(Q^2 + \frac{XQ}{L} \right).} \quad (120)$$

Proof. By (29),

$$|\hat{A}_{\lfloor J \rfloor, q}(0)| \leq |\mathcal{S}_{\text{sh}}^{\text{all}}| + |\mathcal{S}_{\text{sh}}(c_{\alpha, \gamma}(j) > \lfloor J \rfloor)|.$$

Use (114), (30), and (117). The calibrated packet differs by the two drift legs bounded in (31); adding them proves (120). $\square$

This corollary is a direct amplitude estimate. It matches the amplitude budget associated with the 1/400 flatness gate in TPC-32, but it does not prove the flatness-ratio inequality itself because no lower bound for $\sum_n |A_C(n)|^2$ is available.

9 Sharp obstructions and the existing-theorem boundary

The preceding theorems create an average route around pointwise affine cancellation. They do not make a prescribed direction small. We record three exact obstructions before comparing the physical requirement with existing Chowla–Elliott technology.

9.1 Three sharp interface obstructions

Proposition 9.1 (A fixed direction contains two-point Chowla). *The family $sU - aD = h$ contains the shift-one Möbius correlation. In particular, take*

$$a = s = h = 1, \quad x = 1, \quad y = 0. \quad (121)$$

Then

$$D_t = t, \quad U_t = t + 1, \quad \mu(D_t)\mu(U_t) = \mu(t)\mu(t + 1). \quad (122)$$

Thus an unweighted $o(T)$ theorem uniform over every individual affine direction would imply the still open two-point Möbius Chowla statement.

Proof. The Bézout relation is $sx - ay = 1$. Substitution in (36) gives (122). $\square$

The next example shows why separate Fourier pseudorandomness of the two legs cannot control a prescribed pairing.

Proposition 9.2 (Fourier-flat legs with a coherent line). *Let q be an odd prime, fix $a_0, s_0, h \in \mathbb{F}_q^\times$, and let χ be the quadratic character with $\chi(0) = 0$. Define*

$$f(D) = \chi(D), \quad g(U) = \chi\left(a_0^{-1}(s_0U - h)\right). \quad (123)$$

Then $\bar{f} = \bar{g} = 0$, and every nonzero additive Fourier coefficient of either function has modulus $\sqrt{q}$. Nevertheless,

$$\boxed{\mathcal{C}_{a_0, s_0}^{(h)}(f, g) = q - 1.} \quad (124)$$

Proof. Both functions are affine transforms of the quadratic character, so their means vanish and their nonzero Fourier coefficients are quadratic Gauss sums of modulus $\sqrt{q}$. On the selected line $U = s_0^{-1}(a_0D + h)$,

$$g(U) = \chi\left(a_0^{-1}(a_0D + h - h)\right) = \chi(D) = f(D).$$

Hence the correlation is $\sum_D \chi(D)^2 = q - 1$. $\square$

This counterexample concerns an abstract two-leg Fourier interface. It does not assert that the Möbius function possesses the same coherent structure; it proves that one-leg Fourier bounds alone cannot rule it out.

Proposition 9.3 (The single-modulus no-wrap conflict). *On the high- β physical packet, a single prime modulus cannot both turn the modular determinant condition into an exact integer equation on the ambient affine box and provide complete orbit-slope coverage.*

Proof. On a projected physical layer, the slopes are s and $a = \ell j v$, while the $\tilde{d}$ -interval has ambient length $\asymp D/v$. Therefore the determinant expression varies on the scale

$$a \frac{D}{v} = \ell j D \asymp QJ \asymp X. \quad (125)$$

Table 2: Existing affine-correlation inputs and the missing physical interfaces.

Source	Coefficients	Averaging	Boundary for TPC-33
Tao, Cor. 1.5 [7]	fixed nonproportional affine forms	logarithmic window $x/\omega(x) < n \leq x$, $\omega(x) \rightarrow \infty$	applies to each frozen Möbius pair; no stated uniformity for polynomially growing coefficients and not a fixed dyadic window
Matomäki–Radziwiłł–Tao, Thm. 1.6 [4]	coefficients bounded in terms of A , with an A^2 loss	unweighted average plus a free additive-shift average	applies also to Möbius, but the displayed bound yields no saving when $A = X^\theta$; the free shift is not the determinant-constrained family
Frantzikinakis–Host, Thm. 2.6 [1]	fixed independent homogeneous linear forms	a full two-dimensional box	gives mean zero for aperiodic Möbius, but not on the one-dimensional affine slice or for growing forms
Matomäki et al., Cor. 1.11 [5]	fixed distinct dilation coefficients	average over a common dilation	strong precedent for extra slope averaging; the displayed corollary is for Liouville and does not select one prescribed dilation

An elementary no-wrap lift from $sU - aD \equiv h \pmod{q}$ to equality throughout this ambient box requires $q \gg X$. In contrast, complete coverage by the orbit variable j requires $q \lesssim J$; otherwise the rms factor in (105) contains $(q/J)^{1/2}$. At the physical endpoint $J = X^{133/400+o(1)} \ll X$, these requirements are incompatible. $\square$

The proposition rules out only the most direct one-modulus transplantation of [theorem 6.1](#). It does not rule out a multi-modulus circle method, a delta symbol, a spectral decomposition, or a direct integer large-sieve inequality for the restricted slope set.

9.2 What the proved affine-correlation theorems supply

Several strong results control related averages. Their hypotheses do not coincide with the physical column. The comparison in [table 2](#) uses only the theorem forms stated in the cited primary sources. The earlier TPC-23 paper already records the fixed-form logarithmic bridge in the present reduction [10]; it is not claimed again as a new theorem here.

Two further nearby boundaries are worth recording. Teräväinen proves a fixed gap from the trivial bound for certain roots-of-unity-valued multiplicative functions on fixed nonproportional affine forms [8, Thm. 2.6]; this is not a mean-zero theorem, and the roots-of-unity hypothesis does not directly permit μ , which also takes the value zero. Pilatte proves a quantitative logarithmic shift-one result for the Liouville function [6, Thm. 1.1]; it is a preprint result for a particular fixed pair, not a general growing-coefficient dyadic Möbius theorem.

In summary, no cited theorem simultaneously supplies:

- (i) polynomially growing affine coefficients;
- (ii) cancellation on every fixed dyadic shell;
- (iii) averaging along the fixed-determinant constrained family;
- (iv) the actual joint row mask and physical weights; and
- (v) a power ledger strong enough for (113).

These requirements remain an additional analytic input. The projective identity in this paper identifies one viable averaged geometry, but does not claim that existing Chowla–Elliott theorems already close it. The distinction between sign-sensitive prime detection and sign-blind sieve information is the classical parity boundary [2].

10 The high- β ledger and the next gate

We now specialize the exact interfaces to the source-compatible endpoint of TPC-28 [12]. The purpose is to measure what one-fiber cancellation can and cannot accomplish, and to state the next target without implying that it has been proved.

10.1 Physical exponents

The row, orbit, prime-source, and opened-divisor exponents are

$$\beta = \frac{267}{400}, \quad Q = X^{267/400+o(1)}, \quad J = X^{133/400+o(1)}, \quad (126)$$

$$\lambda = \frac{99979}{210000}, \quad \ell = X^{\lambda+o(1)}, \quad d_* = \frac{10049}{52500}, \quad (127)$$

$$D = X^{d_*+o(1)}, \quad T = X^{193/500+o(1)}. \quad (128)$$

The exponent λ here is unrelated to the finite projector coefficient $\lambda_G(v)$ from section 5. Here $Q = LD$ and $QJ \asymp X$. In an unprojected affine column,

$$\ell j = X^{\lambda+133/400+o(1)} = X^{42451/52500+o(1)}. \quad (129)$$

The reflected complement satisfies

$$s \ll X/T = X^{307/500+o(1)}. \quad (130)$$

After the row-gcd projection, the second slope is

$$a = \ell j v, \quad v \mid e, \quad a \leq X^{1+o(1)}, \quad (131)$$

because

$$\frac{42451}{52500} + \frac{10049}{52500} = 1. \quad (132)$$

The corresponding affine fiber has length

$$|I| \ll 1 + X^{d_*+o(1)}/(sv). \quad (133)$$

Thus many slope layers are arithmetically nondegenerate but contain only $O(1)$ points. Nonzero determinant alone does not create a long average.

10.2 Why square root on every long fiber is still insufficient

The longest possible affine fiber has length D , attained only when the product sv is bounded. Suppose, optimistically, that every such fiber could be bounded with square-root cancellation while all outer variables were reassembled absolutely. The largest relative saving visible to that strategy would be

$$D^{-1/2} = X^{-10049/105000+o(1)}. \quad (134)$$

Indeed, if the fiber lengths are $n_i \leq D$, then after absolute reassembly the ratio of square-root bounds to trivial masses is a weighted average of $n_i^{-1/2}$, and hence cannot improve on $D^{-1/2}$. The required amplitude saving is

$$J^{-1} = X^{-133/400+o(1)}. \quad (135)$$

The exact missing exponent is

$$\boxed{\frac{133}{400} - \frac{10049}{105000} = \frac{49727}{210000} = 0.236795\dots} \quad (136)$$

This is a limitation of a specified proof strategy, not a lower bound for the actual Möbius sum. It proves that even a pointwise square-root theorem for each longest fiber cannot close the endpoint if every remaining index is treated by the triangle inequality. The missing gain must use collective averaging across some of

$$\ell, \quad j, \quad s, \quad v, \quad \text{and the opposite row.} \quad (137)$$

10.3 The restricted physical projective-energy problem

The next analytic target is exactly

$$\boxed{E_L + E_R \ll_{\varepsilon, h, \mathcal{D}} X^\varepsilon Q^3,} \quad (138)$$

with E_L, E_R defined in (107) and (109), and with each inner column expanded as in theorem 5.2. The estimate must retain:

- (a) both exact polarizations;
- (b) the actual row coefficients and row-gcd projection;
- (c) the fixed-period and smooth orbit factors;
- (d) the +1 occupancy in (133); and
- (e) the exceptional coefficient pairs allowed by determinant-fiber averaging.

There are several nonexclusive routes to (138):

- (i) prove an integer restricted-projective large sieve directly for $a = \ell j v$ and the physical s -range;
- (ii) use a multi-modulus delta/circle decomposition to separate no-wrap from slope coverage;
- (iii) expose a genuine second long variable and apply a correlation theorem in two variables before taking absolute values; or
- (iv) split long and sparse fibers and combine a TT^* estimate on the long range with an incidence-energy estimate on the $O(1)$ -point range.

Each route requires a new norm and conductor ledger. The present paper does not select one as already successful.

10.4 Endpoint consequences if the gate is proved

At (126),

$$Q^2 = N_0 J^{-1} = N_0 X^{-133/400+o(1)}. \quad (139)$$

Thus (138), together with theorem 8.1, gives every strict saving below 133/400 for the matched raw shell. The large-content correction at $C = J$ has the same Q^2 size.

The drift term is smaller still. From (126)–(127),

$$\frac{XQ/L}{Q^2} = \frac{J}{L} = X^{-15077/105000+o(1)}. \quad (140)$$

Consequently the calibrated shell would also be $O(X^{o(1)}Q^2)$.

Even that conditional endpoint closes only the residual packet inherited from the preceding reductions. It does not supply the independent positivity and main-term reconstruction needed for a prime-pair theorem. The next gate is therefore substantial but sharply localized: it is a restricted projective L^2 estimate, not a disguised assertion that the twin-prime conjecture has already been proved.

11 Exact certificate and claim boundary

The accompanying standard-library certificate is

`experiments/tpc33_certificate.py.`

It uses integer and `Fraction` arithmetic and exception-based checks. No Python `assert` statement, floating-point root of unity, or random input is used. Running

```
python experiments/tpc33_certificate.py
python -O experiments/tpc33_certificate.py
```

regenerates byte-identical JSON and identical standard output in both modes.

11.1 What is checked

The certificate performs 506,900 exact checks, distributed as in [table 3](#).

Table 3: Exact finite checks in the TPC-33 certificate.

Check family	Count
Affine normal form, gcd identities, and Möbius layers	248,914
Reduced and projected affine origins	83,646
Local squarefree root counts	9,853
Actual-mask gcd projector and coefficient mass	135,320
Projective second moment and product-slope lift	714
Quadratic-character coherent-line obstruction	27,790
Finite column Cauchy–Schwarz inequalities	630
High- β rational ledger	9
Scope flags	24
Total	506,900

The affine checks enumerate primitive coefficient triples and verify

$$sU_t - aD_t = h, \quad \begin{pmatrix} h \\ t \end{pmatrix} = \begin{pmatrix} s & -a \\ -y & x \end{pmatrix} \begin{pmatrix} U_t \\ D_t \end{pmatrix}, \quad (141)$$

all four gcd identities, the nonprimitive Möbius layer, and the discriminant h^2 . The local sieve checks enumerate complete residue systems modulo p or p^2 and verify the exact $p - 1$, $p^2 - 1$, and $p^2 - 2$ allowed-class counts.

The mask checks verify (66) and (67) on finite ranges. The projective checks evaluate both sides of (88) directly over finite fields, without approximating additive characters. The coherent-line check uses the exact quadratic-character autocorrelation

$$\sum_x \chi(x)\chi(x+k) = \begin{cases} q-1, & k=0, \\ -1, & k \neq 0, \end{cases} \quad (142)$$

which certifies Fourier magnitude $\sqrt{q}$ through finite-character orthogonality.

11.2 Reproducibility values

The archived run has:

$$\begin{array}{ll} \text{normalized source SHA-256} & \text{a55c3d71c0851870ea0b9bbd1e9ef938} \\ & \text{4e70885e5de1ddc685c309ec1ec12c67,} \end{array} \quad (143)$$

$$\begin{array}{ll} \text{JSON SHA-256} & \text{c7d08c9020590d81bc590a6caa7d0f36} \\ & \text{ad3f553532e580fbd03553638a0b3b5,} \end{array} \quad (144)$$

$$\begin{array}{ll} \text{certificate digest} & \text{e5a28b8dcac36154ff620c2e644eed7f} \\ & \text{d163d00568fcd90d8e4443beafcfe37b.} \end{array} \quad (145)$$

The normalized source hash replaces all line endings by line feeds before hashing. The certificate digest hashes the canonical JSON payload before the digest field is appended.

11.3 What is not certified

Finite enumeration verifies algebra and exact bookkeeping, not an asymptotic analytic estimate. In particular, the certificate does not:

- (i) certify the error term in (57); that term is proved in the text by a CRT sieve and a large-square tail;
- (ii) prove signed cancellation in any growing affine Möbius family;
- (iii) prove the physical Q^3 input (113);
- (iv) prove the zero-frequency flatness ratio of TPC-32;
- (v) close every residual packet in the earlier prime-pair reduction;
- (vi) prove positivity or a Hardy–Littlewood asymptotic;
- (vii) prove infinitely many twin primes; or
- (viii) break the sieve parity barrier.

The proved statements are the integral normal form, its squarefree-support theorem, the actual-mask column disintegration, the finite projective second moment and its corollaries, the exact conditional Q^3 -to- Q^2 transfer, and the sharp obstructions. The remaining mathematical task is (138), together with all later positivity and reconstruction stages. This delineates the proved, conditional, and open parts of the argument.

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